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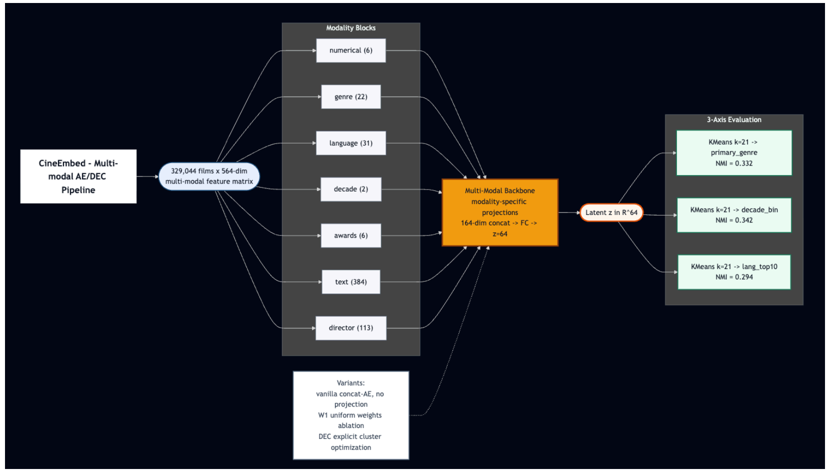
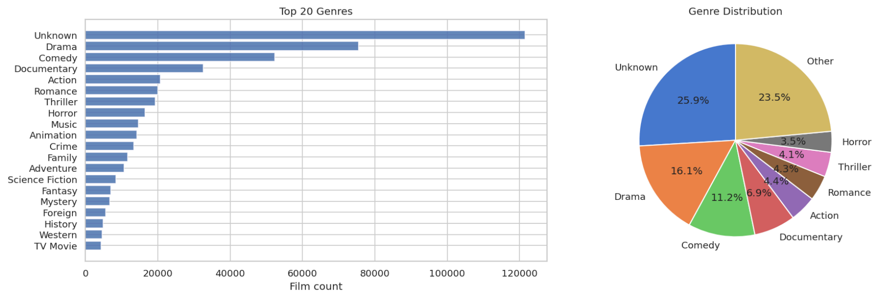


Problem & Motivation

The data is heterogeneous, sparse, and partially missing.

- **329,044 films × 564 features** across **7 modality blocks**
- Genre is **multi-label and long-tailed** — 21 classes, top-3 dominate
- Language is **99% zero per row** (one language per film, 31 indicators)
- 96.8% of films have **no Wikipedia director bio** → masking required
- 7.4% of films have **no release date** → relevant later (Finding 9)

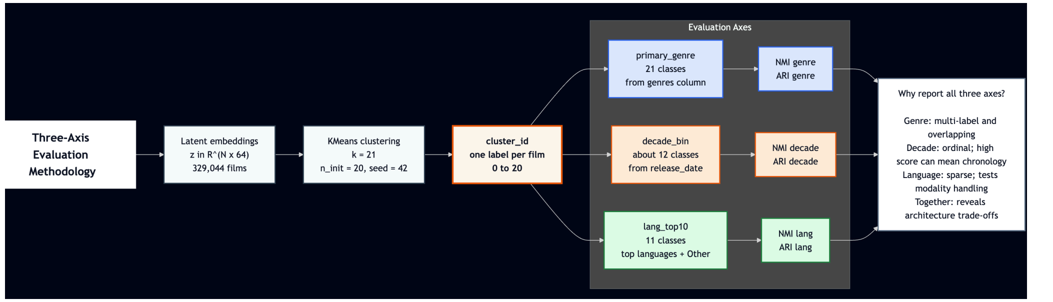


Three-Axis Evaluation, By Design

A **single ground-truth axis** would hide the trade-off this work is about.

We evaluate every model against three orthogonal label axes simultaneously:

- `primary_genre` — 21 classes (multi-label collapsed)
- `decade_bin` — ~12 buckets (1900s through 2020s + missing)
- `lang_top10` — 11 classes (top 10 + "other")



For each axis: cluster the 64-dim latent with **KMeans (k=21, n_init=20)**, then score with

Data Pipeline

Three sources merged → 564-dim matrix.

Block	Dim	Notes
numerical	6	log-popularity, runtime, vote
genre	22	21-way one-hot + has_genre
language	31	top-30 langs (very sparse)
decade	2	decade_norm + has_release_date
awards	6	Oscar/BAFTA/Cannes log-counts
text	384	`all-MiniLM-L6-v2` embeddings
director	113	bio_pca_64 + lang/country/flags

Three-Tier Model Taxonomy

We trained **6 models** spanning three tiers — designed so each tier isolates one source of the deep-vs-classical gap.

Non-deep baselines

- `kmeans_raw_k21` — KMeans on 564-dim
- `pca_kmeans_k21` — PCA-64 + KMeans

Simple deep

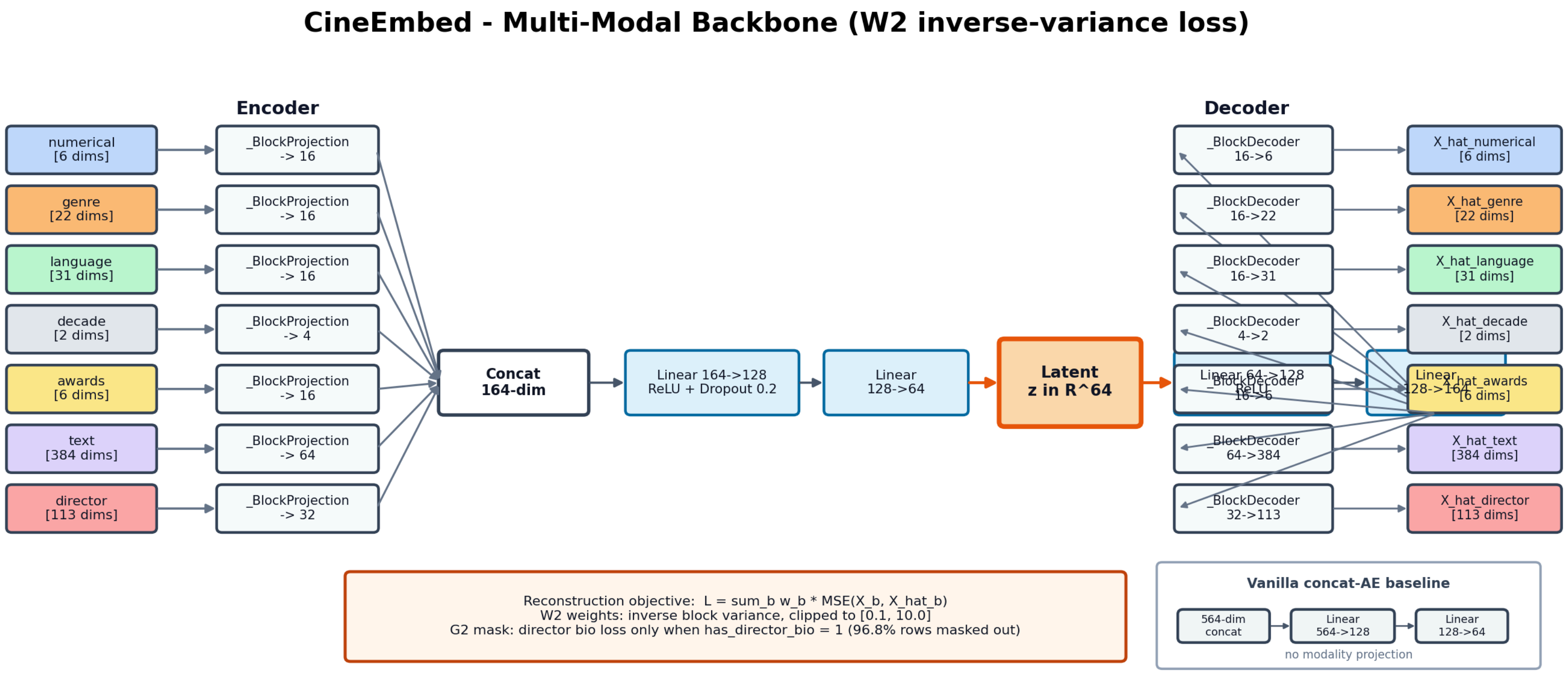
- `vanilla_ae_z64` — concat 564→128→64

Multi-modal deep + ablations

- `ae_z64` — 7 projections + W2 + G2
- `ae_z64_w1` — same arch, **W1 uniform**

Multi-Modal Architecture

7 modality projections → concat (164-dim) → shared backbone → **z** = $\mathbb{R}^{64}$



Loss (W2 inverse-variance, with G2 masking):

Results — Main Comparison Table

All 6 runs, z=64 latent, KMeans k=21:

Tier	Run	genre_NMI	genre_ARI	decade_NMI	decade_ARI
Non-deep	`kmeans_raw_k21`	0.109	0.063	0.233	0.093
Non-deep	`pca_kmeans_k21`	0.084	0.061	0.224	0.085
Simple deep	`vanilla_ae_z64`	0.287	**0.247**	**0.369**	0.175
Ablation (W1)	`ae_z64_w1`	0.165	0.094	0.367	0.176
Multi-	`ae_z64`	0.328	0.229	0.341	**0.211**

Headline: Deep Beats Non-Deep By 3–5×

+205%

DEC `genre_NMI = 0.332` vs `kmeans_raw_k21 = 0.109`

Genre NMI

DEC vs kmeans_raw → **+205%**

DEC vs pca_kmeans → **+295%**

Genre ARI

DEC vs kmeans_raw → **+287%**

DEC vs pca_kmeans → **+300%**

Language NMI

DEC vs pca_kmeans → **+213%**

Multi-modal vs vanilla → **+178%**

→ Pre-registered **H2** (best-deep ≥ 1.10 × best-non-deep): **PASS by an order of**

Architecture Wins on Language (+178%) — and the Pareto Trade-Off

Axis	vanilla	multi-modal	Δ
`lang_NMI`	0.095	0.264	**+178%**
`genre_NMI`	0.287	0.328	+14%
`decade_NMI`	0.369	0.341	−7.6%

Why language wins so big: modality projection allocates capacity to the 31-dim 99%-zero language block.
The vanilla concat-AE is dominated by the 384-dim text block.

Why decade slightly loses: capacity has to come from somewhere — the 2-dim decade signal is "easy", so the multi-modal model can afford to spend less on it.

→ **Principled trade-off, not a bug**

W2 Weighting Is Critical — W1 Collapse Visible

Axis	W2 (`ae_z64`)	W1 (`ae_z64_w1`)	Δ
`genre_NMI`	0.328	0.165	−50%
`lang_NMI`	0.264	0.070	−73%
`decade_NMI`	0.341	0.367	+8%

Without inverse-variance weighting, **high-dim low-variance modalities lose all gradient signal** (text 384, language 31). The 2-dim decade block survives because StandardScaler already gave it variance ≈ 1.

Diagnostic side-effect: W1 early-stops at epoch 37 (vs 69 for W2). Patience exhaustion = "model gave up".

DEC Sharpens Boundaries — and Latent Topology Evolves

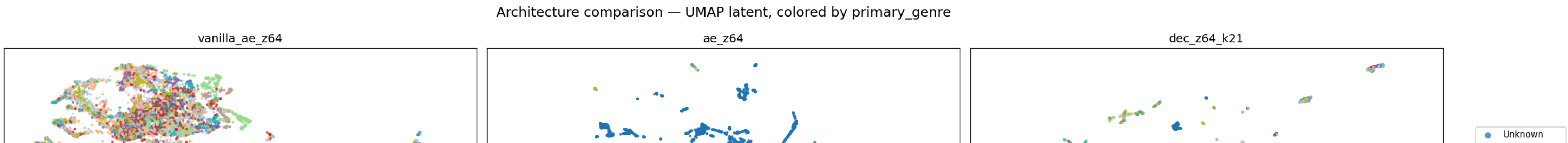
DEC initialized from `ae_z64`, trained 21 epochs of KL+recon:

Metric	AE (init)	DEC	Δ
`genre_NMI`	0.328	**0.332**	+1.2%
`genre_ARI`	0.229	**0.244**	**+6.6%**
`lang_NMI`	0.264	**0.294**	**+11.4%**

ARI gain > NMI gain = DEC tightens partitions, doesn't restructure. Diagnostic signature.

Cluster health: `total_reinit = 0` over 21 epochs.

All 21 KMeans++ centroids survived KL training without collapse.



Discovery: Missing-Data Manifold

In the DEC decade plot, films with `decade_bin = 0` (~7.4%, **missing release_date**) form an **isolated red cluster** in the upper-right.

The model **was not forced** to encode missingness as latent geometry — `has_release_date` is one binary input among 564.

Yet across all four deep architectures, "no release date known" emerges as a **dimension of latent geometry, not just a flag**. DEC compresses it into the cleanest partition.

Not predicted by H1–H3. This is a post-hoc representation-learning interpretability win.

Practical: latent-space queries naturally cluster missing-metadata films — useful for downstream data-quality triage.

Conclusion + Future Work

All three pre-registered hypotheses PASS:

- **H1** — DEC > AE on `genre_NMI`: ✅ 0.332 > 0.328 (+6.6% on ARI)
- **H2** — Best deep > 1.10 × best non-deep: ✅ **PASS by +205%**, an order of magnitude over threshold
- **H3** — Best deep `genre_NMI` > 0.15 floor: ✅ 0.332 ≫ 0.15

Plus a bonus post-hoc finding (missing-data manifold) not predicted by H1–H3.

Best model: `dec_z64_k21` — `genre_NMI=0.332`, `lang_NMI=0.294`, `decade_NMI=0.342`.

Deferred for the final report:

VAE family (z=32, 64, 128) · AE z-dim sweep · F1 (no-text) ablation · F2 (no-director) ablation · DEC k-sweep (9-cell grid) · W4 (Kendall learned uncertainty) · linear probing · 5-seed CIs · full reproducibility audit.